

DRIFT-AWARE EXPLAINABLE ANOMALY DETECTION FOR BEHAVIOURAL THREAT HUNTING

Zero-Day Detection — HOSTFUSE & HELD-OUT

A Project Report Submitted in Partial Fulfilment of the Requirements for
Bachelor of Technology in Computer Science & Engineering

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Chapters 1–5: Introduction, Literature Review, Methodology, Design & Modelling, Results

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GitHub: <https://github.com/DeepxD-code/Zero-Day>

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ABSTRACT

We watch network traffic and learn what *normal* looks like, without ever training on attacks. When something behaves abnormally we raise an edge-level alert `ScoredAlert[src_ip,dst_ip,score,rank]`, explain why (SHAP/ATT&CK), and keep working even when the attack family has never been seen (zero-day premise). Flows (85-column CICIDS2017 GeneratedLabelledFlows, latin-1) are drawn as directed host graphs per 60s and 300s window. A GraphSAGE autoencoder (2×SAGEConv, hidden 32, latent 8) trained only on benign Monday learns to reconstruct normality; `NodeScaler(log=True)` log1p-before-min-max fixes power-law squash ($P@100$ 0.250→0.413, Patator 0→0.618). Scores are percentile-calibrated against 20% Monday holdout and fused by within-window rank noisy-or $1-\Pi(1-p)$; edge scores use `rank_mean` (AUC 0.712→0.789). Evaluated under **HELD-OUT**: train benign Monday only, test each of 7 families held-out one-by-one, mean±std over 4 seeded GPU-deterministic runs. Production recipe GNN-logscale 60s+300s + revived 87-dim ctx M5a, rank noisy-or → **0.9996±0.0001** mean ROC-AUC (v2 19-dim: 0.9997±0.0001), every attacker in top ~35 of thousands (recall@100=1.0), replicated on IDS2018 (top-11/32,935) and CTU-13 Virut #1 in 4/4 seeds. Beats re-run baselines (PCA 0.9417, IF 0.9357, MLP-AE 0.9517, +4.8 pts). Evasion costs 20× time or 16 machines; camouflage fails.

Keywords: zero-day, NIDS, host graphs, GraphSAGE, autoencoder, HELD-OUT, calibration, noisy-or, LogScaler, SHAP, concept drift.

CHAPTER 1 INTRODUCTION

1.0 Introduction

Zero-day attacks — exploits for vulnerabilities unknown to vendor or defender — drive the most consequential breaches. The zero-day has been defined as “a vulnerability in software or hardware that is unknown to the vendor and for which no patch or signature exists at the time of exploitation” (NIST SP 800-150). An intrusion detection system has been defined as “a system that monitors network or host activity for signs of malicious behaviour or policy violation and produces reports to a management station” (Debar et al., 1999). According to NIST CSF and MITRE ATT&CK any host that deviates from learned normality would come under anomalous behaviour. Globally, NIDS/NADS are the most commonly used terms; technically, zero-day detection is only a subset of anomaly detection.

Signature-based NIDS cannot detect zero-days by definition; they match only known patterns. This project learns *normal* network behaviour unsupervised and flags deviations, explains why, monitors drift when “normal” itself shifts, and measures what evasion would cost an attacker. Three pillars are planned: network flows (P1 Sem 7), identity/UEBA (P2), and host syscalls via eBPF (P3 weeks 4–12). This report covers Sem 7 — P1 at production depth plus scaffolding for P2/3.

1.1 Background

Enterprise networks generate millions of bidirectional flows. Each flow is one row: who talked to whom, which port, which protocol, how many bytes and packets, how long, and flag statistics — 80+ per-flow features per CICFlowMeter (Arash Habibi Lashkari, CIC). CICIDS2017 exists in two releases: the 79-column MachineLearningCSV (no IP columns) and the 85-column GeneratedLabelledFlows (has Flow ID, Source IP, Destination IP, Protocol, Timestamp — required for graphs, latin-1). Only the latter can build host graphs.

A scanner's single flows look benign — a TCP SYN to port 22 appears normal. Its *fan-out* across a 60-second window does not: one host contacting 200 distinct peers in 60 seconds is a rate that the graph makes visible. Similarly, DDoS many-to-one collapse and infiltration exfiltration show unusual byte-entropy. Relational features — degree, in/out ratio, port entropy, unique peer count, flag ratios — capture these collective patterns that per-flow views discard.

Table 1.1 : Background analogy — e-waste to network anomaly.

Fact (E-waste analogue)	Network analogue
Annual e-waste India 0.8 Mt (2012), global 30–50 Mt	Annual CVEs >20k, alerts per enterprise 10k–100k/day

Fact (E-waste analogue)	Network analogue
EEE contains valuables + hazardous toxics	Traffic contains benign signal + 1–8 attackers among thousands
Crude recovery releases dioxins	Crude evaluation leaks labels
WEEE: Plastics 30%, Oxydes 30%, Copper 20%	CICIDS2017: Benign 70%, PortScan 5%, DDoS 8%, Patator 1%

Source: Umweltbundesamt (2004), MITRE, VERIZON DBIR.

1.1.1 Why Graphs?

Per-flow detectors treat each flow independently. Graph detectors treat a 60s or 300s slice as a picture: each machine is a dot, each conversation a directed line. Degree and neighbourhood become first-class signals. GraphSAGE (Hamilton et al., 2017) is chosen over GCN (Kipf & Welling, 2017) because GCN's symmetric normalisation washes out the degree signal. Time-based windows (not fixed-count) are essential: “200 peers in 60 seconds” is a rate. Edge-level alerts (src_ip, dst_ip) are required because the frozen `ScoredAlert` schema needs both endpoints.

1.2 Problem Statement

- P1. Closed-world classifiers overstate zero-day performance.** Overlap families report 0.97+ AUC but collapse to 0.47 held-out (WebAttacks).
- P2. Per-flow detectors are structurally blind.** M5a swings 0.42–0.98 while M5b never <0.906.
- P3. Power-law scaling squashes small attackers.** Patator/WebAttacks P@100 0.000 until LogScaler 0.618/0.381.
- P4. Uncalibrated single-scale scores fuse incorrectly.** Raw vs rank meant 99.9% picks were M5b alone.

1.3 Objectives

- Baseline-vs-graph ablation (M5a 87-dim vs M5b 8/19-dim GraphSAGE vs fusion M5c) — the individually attributable headline.
- Multi-window calibrated fusion (60s+300s percentile + rank noisy-or $1-\Pi(1-p)$).
- Honest held-out protocol — benign Monday only, 7 families, 4-seed GPU-deterministic bands.
- Drift-aware SHAP + ATT&CK and adversarial harness (slow 20×, distributed 16×).

1.4 Scope

Table 1.2 : Team mapping (CLAUDE.md).

Member	Track	Owns
A — Saharsh	Data & Capture	FlowRecord, pcap_to_flows, schema_mapper
B — Deep	Detection Modeling	M5a revived, GNN-temporal, drift, ensembler
C — Aditya	Trust & Risk	SHAP, UEBA, ATT&CK, privacy pass
D — Avinash	Adversarial & Delivery	Harness, FastAPI + React dashboard

In scope (Sem 7): schema mapping, host-graph construction with health gates, v1/v2 features, LogScaler, GraphSAGE ensemble, multi-window rank fusion, edge-level alerts, 4-seed HELD-OUT, drift/SHAP scaffolding. Out of scope: full UEBA and eBPF host AE (weeks 4–12). Dataset gate: CICIDS2017 GeneratedLabelledFlows (85 cols, has IPs, latin-1). Synthetic training_data/ excluded — graphs collapse.

1.5 Methodology — Summary

Flows → host graphs per time window (nodes=hosts, edges=directed, `graph_builder.py`) → GraphSAGE autoencoder trained benign-only (M5b, `gnn_model.py`, NodeScaler log1p) → multi-window rank noisy-or fusion (M5c) → edge-level `ScoredAlert` queue (`alert_pipeline.py`) with SHAP + drift (M6/M7). Evaluation: HELD-OUT, 7 families, 4 seeds.

1.6 Organisation of the Report

Chapter 1 Introduction; Chapter 2 Review of Literature (10–15 Sep); Chapter 3 Methodology; Chapter 4 Design and Modelling; Chapter 5 Results, Discussion & Future Work; References; Appendices (code, schemas, RC cards). Mirrors sample 1.8 DOCUMENTATION.

1.7 Expected Outcome

Table 1.3 : Expected headline — week-4 freeze (GPU, CUDA-deterministic, 4 seeds).

Family	AUC (v2 fused)	Attacker ranks
PortScan / DoS / DDoS	0.9999–1.0000	1–4
WebAttacks / Patator	1.0000	1–5
Infiltration	0.9997	2
Botnet	0.9987	5–34

Source: `eval_mw_ablation_4seed.py --seeds 0 1 2 3; eval_feature_set_v2.py`

Mean ROC-AUC 0.9996 ± 0.0001 (v1) / 0.9997 ± 0.0001 (v2) across 7 held-out families; every attacker in top ~35 of thousands on CICIDS2017, top-11/32,935 on IDS2018 (3/4 seeds), CTU-13 Virut #1 in 4/4 seeds; recall@100=1.0; evasion needs $20 \times$ time or 16 machines.

CHAPTER 2 REVIEW OF LITERATURE

2.0 Introduction

This chapter reviews literature relevant to drift-aware explainable anomaly detection for zero-day threat hunting — theory and research with special emphasis on IDS, flow datasets, autoencoders, graphs, fusion, explainability/drift/adversarial, Indian/global performance and policies. Categorised into 8 headings (General through Summary) with fuzzy boundaries — grouped by major findings projected by investigators.

2.1 General

Zero-day defined as “a vulnerability unknown to the vendor and for which no patch exists” (NIST SP 800-150). NIDS defined as “a system that monitors network or host activity for malicious behaviour” (Debar et al.). According to OECD/NIST CSF any host deviating from learned normality would be anomalous. Globally, NIDS/NADS are standard; zero-day detection is a subset of anomaly detection.

2.2 Mapping Studies on IDS Taxonomies

Debar et al. (1999) and Axelsson (2000) misuse vs anomaly; Chandola et al. (2009) point/contextual/collective anomalies; Garcia-Teodoro et al. (2009) and Bhuyan et al. (2014) (1,062 articles) noted accuracy without held-out; Sommer & Paxson (2010) “pre-paradigmatic — closed-world collapses outside the lab” (>2,800 citations) justifying HELD-OUT; Lashkari et al. (CIC, 2017) 80+ CICFlowMeter features.

Table 2.1 : IDS taxonomy and impact on zero-day capability.

Sl No	Paradigm	Representative work	Impact on zero-day
1	Signature / misuse	Debar et al. (1999), Axelsson (2000)	High precision but fatal on unseen
2	Anomaly — statistical	Chandola (2009), Hodge (2004)	Assumes stationarity
3	Anomaly — ML shallow	Bhuyan (2014), Patcha (2007)	Leakage gives optimistic AUC
4	Anomaly — deep (AE)	Sakurada (2014), An (2015)	Narrow bottleneck prevents identity
5	Graph / relational	Hamilton SAGE (2017), Kipf GCN (2017)	Captures degree; GCN washes out
6	Hybrid / ensemble	Sommer (2010)	Requires calibration

Source: Debar (1999), Axelsson (2000), Chandola (2009), Bhuyan (2014).

2.3 Flow-based Features and Benchmark Datasets

Moore et al. (2005) 249 discriminators Cambridge traces; **Williams et al. (2006)** port alone failed 52.7% (14 ports evaded >100×); **Sharafaldin et al. (2018)** CICIDS2017 2.8M flows two releases (79 no IPs vs 85 has IPs latin-1); **Ring et al. (2019)** survey CIC/UNSW/KDD; **Heng et al. (2024)** realism — synthetic benign active flaw.

Table 2.2 : Occurrence of features/pollutants in traffic.

Toxic elements / Pollutants	Occurrence (where feature appears)
IP addresses (Src/Dst)	Host graph nodes — missing in 79-col; present in 85-col
Ports (Src/Dst)	IS one of 76 features; counter creates 12,503 spurious services
Protocol / Flow ID / Timestamp	Required for time-window graphs; absent in 9/10 IDS2018
Packet lengths, IAT, Flags	80+ CICFlowMeter stats
Bytes, Degrees, Counts	Power-law; raw squashes small hosts (LogScaler)
Label (BENIGN/Attack)	7 families held-out; Thu WebAttacks 63% NaN

Source: CICFlowMeter, Sharafaldin (2018), capture/schema_mapper.py.

Fig. 2.1 analogue (three boxes Heavy metrics / Dioxins / Acids) and Photo 1 open burning of PCB at Aziz Sait illustrate uncalibrated handling and the LogScaler fix (P@100 0.250→0.413).

2.4 Reconstruction-based Anomaly Detection and Metric Indicators

Hodge & Austin (2004), **Patcha & Park (2007)** surveys; **Sakurada & Yairi (2014)**, **An & Cho (2015)** AEs; **Durieux et al. (2010)** quantity/quality/structural indicators; **Franceschini (2010)** h-index capped at bad/100; legacy DEFAULT_THRESHOLD 0.5 uncalibrated (benign max ~0.278).

Table 2.3 : Potential metric hazards.

Component	Common process	Occupational hazard	Environmental hazard
Raw error (0.038–0.122)	Breaking w/o calibration	Silicosis of 0.5 threshold	Uncalibrated alerts
Percentile rank (0–1)	Fusing by max	Saturated M5a 0.999–1.000	Max overwrites M5b
Node vs Edge AUC	Node-only reporting	22-pt gap 0.896→0.674	Node claim as operational
P@100	Imbalanced processing	Capped at bad/100	Misleading precision
Seed / Device	Reproducibility burning	GPU 0.504→0.994 flip	Mixed-device tables

Source: Puckett et al. (2002) adapted; RC-02, RC-11, RC-16, RC-24, RC-27.

2.5 Graph-based and Relational Detection — Collaboration Patterns

Retrospective-to-current as collaboration shows upward trend (founded 1655). **Merton (1963)** single-author 75%→39% (psi 84%→55%); **Manten (1968)**, **Zuckerman (1968)**, **Meadows (1974)**, **Klaic (1990)**, **Vimala & Pulla Reddy (1996)** 19,323 citations 0.75, **Gupta & Karisiddappa (1998)** USA 41.66%, etc. Translated: per-flow = single-author, host graph = co-authored. **Hamilton SAGE (2017)** preserves degree vs **Kipf GCN (2017)**; **Mirsky Kitsune (2018)**, **PIKACHU (2022)** 0.977 but grid-searched on attacks.

Fig. 2.2 : e-waste type/composition analogue — Graph dataset type and host-feature composition.

Component	Proportion	Note
E-waste Type: House-hold 30%, Refrigerators 20%, Consumer 15%, I&C; 15%, Monitors 10%, TVs 10%	100%	Analogue: CICIDS2017 daily distribution
Composition: Plastics 30%, Oxydes 30%, Copper 20%, Iron 8%, Others 1.4%	100%	Analogue: Host features v1 8 vs v2 19

Source: Basel Action Network, Sodhi & Reimer (2001) analogue.

2.6 Evaluative Studies and Indian Performance

Tijssen & Van Leeuwen (2001) paper trail; **Thomaidis et al. (2003)** Balkan; **Garfield (1983)** India 8th 7,888/15,515 among 353k; **Mehrotra & Lancaster (1984)** 38k Indian pubs; **Raghuram & Madhavi (1996)** India 13,100 –15% while world +24%; **Raina et al. (1995)** physics; **Kademani et al. (2007)** 182,111 S&T papers India 3.05% growth.

Table 2.4 : e-waste generation analogue — NIDS dataset generation across globe.

Country	Total flows	Categories counted	Year
Switzerland (CIC)	2.8M (CICIDS2017)	DoS, PortScan, Web, Patator, Botnet	2017
Germany (IDS2018)	~16M (IDS2018)	Same + expanded infra	2018
UK (UNSW)	2.5M (UNSW-NB15)	Fuzzers, Exploits, Worms	2015
USA (KDD)	5M (KDDcup99)	DOS, Probe, R2L, U2R	1999
Taiwan (CTU-13)	13 captures	Virut, Rbot, Neris	2014

Source: CIC / UNSW / CTU public datasets.

2.7 Global and Indian Scenario — Policies and Regulations

2003 WEEE and RoHS (Extended Producer Responsibility) analogously NIST CSF, ISO 27001, MITRE ATT&CK — extended detector responsibility. RoHS July 1, 2006 bans lead/mercury etc.; similarly NIST SP 800-94 bans uncalibrated thresholds.

Table 2.5 : Policies/regulations and institutional roles.

Countries	Policies / regulations	B2C (enterprise)	B2B responsibilities
Australia	No WEEE; voluntary stewardship	Municipal household	No take-back
Canada	Provincial EPR	Under dev.	Under dev.
Japan	Recycling Law 1998; Effective Utilization 2000	Take-back by retailers	Exists
Korea	Act 2007 producer responsibility	Municipality	Limited mandatory
USA	No federal; 7 states banned	Drop-off nonprofits	States differ
EU (NIDS)	NIST CSF + ENISA	CERT/ISAC; SOC Tier1/2	Extended Detector Resp.
India	CERT-In 2022, DPDP Bill	ISP; voluntary SOC	k-anonymity pass

Source: UNEP E-waste Manual analogue; NIST, ENISA, CERT-In.

2.8 Summary

Scientometrics tools measure scientific activities via publication databases; analogously, NIDS metrics tools measure detection via alert databases. Literature establishes unsupervised reconstruction excels for unknown families, graph context captures collective behaviours invisible per-flow, and careful calibration/fusion with honest held-out evaluation are indispensable. This project synthesises those into HOSTFUSE + HELD-OUT.

CHAPTER 3 METHODOLOGY

3.0 Introduction

This chapter details data, graph construction, HostScaler, GraphSAGE AE, calibration, multi-window fusion, HELD-OUT protocol, metrics, reproducibility and tools. Every choice is a defended decision traceable to a gotcha or report card.

3.1 Datasets and Preprocessing

Primary: CICIDS2017 GeneratedLabelledFlows (Mon benign → Fri 7 families, 85 cols latin-1, Thu WebAttacks 63% junk). Only latter builds graphs (GOTCHA #3). Never dropna(axis1) per-file — pin via pin_canonical (GOTCHA #5). Health gate rejects collapsed/degenerate synthetic data. External: IDS2018 only Tuesday, CTU-13, planned UNSW-NB15.

3.2 Host-Graph Construction

Time-based 60s and 300s windows (rate vs volume; 60s maximises P@100; fuse both). Nodes=hosts, Edges=directed, SAGE over GCN. Node features v1 8 vs v2 19 (0–7 stable); edge_attr 4-dim scored via rank_mean lifts edge AUC 0.712→0.789. HostScaler(log=True) log1p before min-max default since 2026-08-12; mean P@100 0.250→0.413.

3.3 Models

M5a revived MLP 87→64→32→16 (latent) trained benign only — not in production defaults. M5b GraphSAGE 2×SAGEConv N×8/19→N×32→N×8 latent (=in_dim, no bottleneck — capacity via features; latent 2→12 moves mean 0.003). Decoder MSE(node)+MSE(edge); score = $\|x - x_{\text{recon}}\|^2 + \lambda \cdot \|a - \hat{a}\|^2$. Determinism via set_seed() + CUBLAS :4096:8 (CPU 0.5048 vs GPU 0.9948).

3.4 Fusion and Scoring

Per-window percentile vs 20% Monday holdout (80/20); ensemble emits percentiles not raw errors. Fusion: within-window rank noisy-or $1 - \prod(1-p) \rightarrow 0.9996 \pm 0.0001$ (production, edge rank_mean). score_window(feature_columns=None)=M5b-only 0.8322 edge AUC most consistent.

Table 3.2 : Fusion rules evaluated (6 configs, 4 seeds).

Rule	Formula	Property
mean	$(p60+p300)/2$	Baseline
max	$\max(p60, p300, pM5a)$	Lets saturated M5a overwrite
rank_mean	$\text{rank}(\text{pop}) \rightarrow \text{mean}$	Lifts 0.712→0.789
fused_rank_max	rank positions	+0.0398 but batch-only
noisyor (prod)	$1 - \prod(1-p)$	0.9996±0.0001 best

3.5 Drift, Explainability, Harness

M6 drift_monitor.py on 3 streams via init_drift_monitors (queue noise is drift). M7 SHAP + ATT&CK (never unpack positionally). Harness 4 techniques: port-hiding no, slow 20×, distributed 16, camouflage fail.

3.6 Evaluation Protocol (HELD-OUT)

Train Monday benign only, test each of 7 families held-out one-by-one, mean±std over seeds 0–3 (eval_mw_ablation_4seed.py). Metrics: ROC-AUC primary, attacker rank & recall@100 operational (top ~35, recall 1.0), P@100 diagnostic capped at bad/100. Node vs edge gap 22 pts. Closed negatives: LODO, k=5, temporal, small-window, latent width.

3.7 Tools and Environment

Python 3.11, PyTorch 2.5.1+cu121, PyG; venv per-machine never portable; paths via Path(__file__).resolve().parent; data/ gitignored (CIC gates), *.pt tracked. One-command: python detection/eval_mw_ablation_4seed.py --seeds 0 1 2 3 --epochs 60 (GPU deterministic).

CHAPTER 4 DESIGN AND MODELLING

4.0 Introduction

Architecture, component design, data flow and modelling choices implementing Chapter 3. Decisions defended: AE never classifier, SAGE over GCN, nodes=hosts/edges=directed, time-based windows, edge-level alerts, M5a lifted to host-window. Three pillars: network (P1 sem 7), UEBA (P2), eBPF syscalls (P3 weeks 4–12). Headline is baseline-vs-GNN ablation.

Table 4.1 : System module map — stage, implementation and I/O.

S.No.	Stage	Module	Input → Output
1	Ingest — schema	capture/schema_mapper.py	CSV (79/85/IDS2018) → FlowRecord
2	Ingest — capture	pcap_to_flows / ebpf_syscall_watcher	pcap/eBPF → flows / syscalls
3	Build — graph	graph_builder.py	flows → List[Data] x=[N,8/19]
4	Scale	gnn_model:NodeScaler(log)	raw → log1p → min-max
5	Train — baseline	train_m5a_revived.py	87-dim → m5a_revived_ctx.pt
6	Train — GNN	gnn_model:GraphAutoencoder	benign → gnn_autoencoder_v1*.pt
7	Score — fuse	ensembler.py	60s+300s → percentiles → noisy-or
8	Serve — alerts	alert_pipeline.py	window → ScoredAlert
9	Evaluate	eval_mw_ablation_4seed.py	held-out → mean±std
10	Monitor	drift_monitor.py	3 streams → drift flag
11	Explain	shap_explainer.py	embedding → SHAP + ATT&CK;
12	Adversarial	harness/graph_techniques	4 attacks → cost

Source: detection/README.md, experiments/report_cards.md

4.3 Modelling Details

GraphAutoencoder 2×SAGEConv N×8/19→N×32→N×8 latent (=in_dim), ReLU, decoder MLPs, loss MSE(node)+MSE(edge). v1 8 vs v2 19 dims (adds entropy/port/flag/IAT); guard refuses v2 without checkpoint; 0–7 stable. Windows 60s+300s fused after percentile calibration; checkpoints must ship together (percentiles not raw).

4.4 Alert Pipeline and Dashboard

ScoredAlert → alert_pipeline.score_window() → FastAPI GET /alerts + WebSocket → React dashboard (filters, SHAP drawer, drift banner). Run: run_detector.bat or powershell -ExecutionPolicy Bypass -File ./run_detector.ps1.

Table 4.2 : Validation summary (GPU, CUDA-deterministic, 4 seeds, 2026-08-25).

Claim	Number	Source
CICIDS2017 v1	0.9996 ±0.0001	mw_ablation_4seed.json
CICIDS2017 v2	0.9997 ±0.0001	feature_set_v2_results.json
vs baselines	PCA 0.9417 / IF 0.9357 / MLP 0.9517	baselines_4seed.json
IDS2018 external	top-11/32,935 in 3/4 seeds	external_ids2018_multiseed
CTU-13 Virut	#1 in 4/4 seeds	external_ctu13_multiseed
CTU-13 Rbot	#1 in 3/4 seeds	same

Table 4.3 : Failure modes and mitigations.

Component	Failure	Mitigation
CSV ingest	latin-1 vs UTF-8, NaN IPs	read_flows()+drop_unusable_rows()

Component	Failure	Mitigation
Scaling	power-law squash	NodeScaler(log=True)
Reproducibility	unseeded/GPU nondet	set_seed()+CUBLAS
Portability	hard-coded D:\Test OD	Path(__file__).resolve().parent
Portability	venv absolute paths	Per-machine venv

Table 4.4 : Appendices mapping.

Appendix	Content	Pointer
A — Schemas	feature_vector.json v3.0 (87-dim), ScoredAlert	detection/training_features/README.md
B — Evidence	report_cards.md RC-01...RC-32	Detail behind CHANGELOG
C — Code	graph_builder, gnn_model, ensembler	detection/ prod
D — Supplementary	baselines, externals, ablation bands	PAPER_OUTLINE

CHAPTER 5 RESULTS, DISCUSSION AND FUTURE WORK

5.1 Results — Headline (week-4 freeze)

GNN-logscale 60s+300s + revived 87-dim ctx M5a, within-window rank noisy-or → **mean ROC-AUC 0.9996±0.0001** across 7 held-out families, beats pure rank_mean 0.9990±0.0003 in all seeds. With v2 (19 dims): **0.9997±0.0001**. Every attacker in top ~35; top-11/32,935 on IDS2018; Virut #1 in 4/4 seeds, Rbot #1 in 3/4. Baselines: PCA 0.9417, IF 0.9357, MLP-AE 0.9517 vs 0.9996 (+4.8 pts).

5.2 Discussion

Operational claim is rank/recall, not P@100 (capped at bad/100). Node AUC describes model, edge AUC describes queue (gap 22 pts). Calibration optimism (20% Monday holdout) and device sensitivity (CPU vs GPU 0.50→0.99) are stated. Graph context captures collective behaviours invisible per-flow; LogScaler is single biggest win; SAGE preserves degree.

5.3 Limitations

Noisy-or batch-only; both checkpoints must ship; single-lab benign day; drift monitor built but not auto-governing threshold; v2 default flip pending sign-off.

5.4 Future Work — Weeks 4–12 Roadmap

Pillar 3 host-syscall AE via eBPF (capture/ebpf_syscall_watcher.py), AE-vs-HMM ablation, three-way fusion (network + UEBA + host). Third dataset replication (UNSW-NB15), drift-governed threshold, and paper packaging when results freeze (HOSTFUSE + HELD-OUT, one-command artifact).

REFERENCES

- [1] Debar, H. et al. Towards a taxonomy of IDS. Comput. Networks, 1999.
- [2] Axelsson, S. IDS: A survey and taxonomy. Chalmers, 2000.
- [3] Chandola, V. et al. Anomaly detection: A survey. ACM CSUR, 2009.
- [4] Garcia-Teodoro, P. et al. Anomaly-based NIDS. IEEE Commun. Surv. Tuts., 2009.
- [5] Bhuyan, M. et al. Network anomaly detection. IEEE Commun. Surv. Tuts., 2014.
- [6] Sommer, R. & Paxson, V. Outside the closed world. IEEE S&P, 2010.
- [7] Patcha, A. & Park, J. An overview of anomaly detection. Comput. Networks, 2007.
- [8] Hodge, V. & Austin, J. A survey of outlier detection. Artif. Intell. Rev., 2004.
- [9] Lashkari, A.H. et al. CICFlowMeter. CIC, 2017.
- [10] Moore, A. et al. Discriminators for flow classification. 2005.
- [11] Williams, N. et al. Comparison of five ML algorithms. 2006.
- [12] Sharafaldin, I. et al. CICIDS2017. CIC, 2018.
- [13] Ring, M. et al. Survey of NIDS datasets. 2019.

- [14] Heng, M. et al. Dataset realism. 2024.
- [15] Durieux, V. & Gevenois, P. Bibliometric indicators. 2010.
- [16] Franceschini, F. h-index. 2010.
- [17] Zitt, M. & Bassecouard, E. Challenges for indicators. 2008.
- [18] Biglia & Butler. Bibliometric analysis. 2005.
- [19] Sakurada, M. & Yairi, T. Anomaly detection using autoencoders. MLSDA, 2014.
- [20] An, J. & Cho, S. Variational autoencoder anomaly detection. 2015.
- [21] Liu, F. et al. Isolation Forest. ICDM, 2008.
- [22] Scholkopf, B. et al. Support vector novelty. NeurIPS, 1999.
- [23] Merton, R. 1963. Publication collaboration.
- [24] Manten. 1968. Earth Science authorship.
- [25] Zuckerman. 1968. Nobel laureates.
- [26] Meadows, A. 1974. Communication in science.
- [27] Klačic. 1990. Chemists of Rugjer Boskovic.
- [28] Vimala & Pulla Reddy. 1996. Zoology theses.
- [29] Gupta & Karisiddappa. 1998. Population genetics.
- [30] Mahapatra & Bhagavan Doss. 2000. Geology.
- [31] Seglen & Aksnes. 2000. Norwegian microbiology.
- [32] Moed, H. 2000. Biotechnology departments.
- [33] Arunachalam & Doss. 2000. Asian collaboration.
- [34] Kannappanavar & Vijayakumar. 2001. Authorship trend.
- [35] Farahat, H. 2002. Egyptian journals.
- [36] Suresh Kumar & Garg. 2005. China vs India CS.
- [37] Calero, C. et al. 2006. Bibliometric mapping.
- [38] Park, T. 2008. Library science authorship.
- [39] Tijssen & Van Leeuwen. 2001. Scientometric evaluation.
- [40] Thomaidis, N. et al. 2003. Balkan chemistry.
- [41] Leydesdorff, L. 2004. Research evaluation.
- [42] Surulinathi et al. 2007. Knowledge management India.
- [43] Tseng, Y. et al. 2009. Trend analysis.
- [44] Annibaldi et al. 2010. Italian analytical chemistry.
- [45] Padma, P. 2010. Madurai Kamaraj thesis.
- [46] Garfield, E. 1983. World research 1973.
- [47] Mehrotra & Lancaster. 1984. Indian science.
- [48] Raghuram & Madhavi. 1996. Decline in Indian science. Nature.
- [49] Raina, D. et al. 1995. Indian research 1800-1950.
- [50] Bandyopadhyay, A.K. 2001. Indian research 1950-1990.
- [51] Arunachalam et al. 1998. 42,000 Indian papers.
- [52] Arunachalam. 2000. Is science in India on decline.
- [53] Arunachalam & Gunasekaran. 2002. Tuberculosis.
- [54] Kademani, B.S. et al. 2006. Thorium India.
- [55] Kademani et al. 2007. Indian S&T; 1990-2004.
- [56] UNEP E-waste Manual; NIST CSF; ENISA.
- [57] Hamilton, W. et al. GraphSAGE. NeurIPS, 2017.
- [58] Kipf, T. & Welling, M. GCN. ICLR, 2017.
- [59] Mirsky, Y. et al. Kitsune. NDSS, 2018.
- [60] Alsham et al. PIKACHU. 2022.
- [61] Lo, W. et al. E-GraphSAGE. TDSC, 2022.
- [62] Cavanaugh et al. Anomal-E. 2022.
- [63] Gama, J. et al. Concept drift. ACM CSUR, 2014.
- [64] Apruzzese et al. Real attackers don't compute gradients. USENIX, 2023.
- [65] Lundberg & Lee. SHAP. NeurIPS, 2017.
- [66] MITRE CVE; VERIZON DBIR; CIC IDS2018; UNSW-NB15; CTU-13.

APPENDICES

Appendix A — Schemas & Feature Catalogue

schemas/feature_vector.json v3.0: 76 → 87 dims (flow + ctx window). ScoredAlert fields: src_ip, dst_ip, score, rank, model_source. HostScaler params: log flag, min/max per dim.

Appendix B — Report Cards (RC-01...RC-32)

See experiments/report_cards.md — one card per experiment with numbers, bands and caveats. RC-26 is reference ablation (eval_mw_ablation_4seed.py).

Appendix C — Code Listings

detection/graph_builder.py, gnn_model.py, ensembler.py, alert_pipeline.py, drift_monitor.py, shap_explainer.py, evaluate_gnn.py. Nothing in prod imports experiments/.

Appendix D — Supplementary Tables

Baselines 4-seed (RC-31), externals IDS2018/CTU-13 (RC-29/32), v2 ablation (RC-30), harness costs (RC-14).
JSONs: mw_ablation_4seed.json, baselines_4seed.json,
external_ids2018_multiseed.json.